

Zone 2 Methodology Concern and Proposed Fix for SIMON

Purpose. I want to discuss a possible methodology issue in my current Zone 2 force-correction design, and a cleaner proposed fix. Zone 2 is the close-encounter overlap region where adaptive sub-stepping is active and the scalar correction can also be invoked.

1. What the current Zone 2 method does

In the current SIMON implementation, pairwise force direction is analytic and only the scalar magnitude is modified. For close pairs, the code first forms a softened distance and softened force:

$$r_{\text{soft}} = \sqrt{r^2 + \epsilon^2}, F_{\text{soft}} = G m_i m_j r_{ij} / r_{\text{soft}}^3$$

The scalar network is trained on the analytic correction target:

$$c = (r_{\text{soft}} / r)^3, \log c = 3 \log(r_{\text{soft}} / r)$$

The applied close-pair force is then $c F_{\text{soft}}$. Algebraically, if c is exact, this cancels the softening and recovers the Newtonian force:

$$c F_{\text{soft}} = (r_{\text{soft}}/r)^3 * G m_i m_j r_{ij} / r_{\text{soft}}^3 = G m_i m_j r_{ij} / r^3 = F_{\text{Newtonian}}$$

Therefore, in Zone 2 the current system is effectively using Newtonian gravity, but it gets there through a softened-force-plus-correction route.

2. Why this is a concern

The concern is not that the current result is numerically invalid. The concern is methodological: the NN is learning a known analytic formula. Since the correction target is exactly available, the same final force can be obtained more directly by computing Newtonian gravity in this region. In other words, for Zone 2 the NN is not learning a new dynamical correction; it is approximating a formula that mainly restores the softened force back to Newtonian.

This also explains the existing ablation result. When the NN is disabled by setting $c = 1$, the applied close-pair force is not Newtonian; it is the uncorrected softened force. The observed accuracy gain from the NN therefore shows that corrected softening is better than leaving the force softened. It does not prove that the NN correction is better than direct Newtonian force.

3. Proposed Zone 2 fix

Proposal: in Zone 2, replace the current $c F_{\text{soft}}$ path with direct Newtonian force. The numerical difficulty of close encounters in this region should be handled by adaptive sub-stepping, while the force law itself should remain exact.

The proposed Zone 2 rule would be:

$$\text{For } 4 \times 10^{-4} \text{ AU} < r < 0.05 \text{ AU: } F_{\text{Zone2}} = F_{\text{Newtonian}}, \text{ with adaptive sub-stepping active.}$$

This is cleaner because Zone 2 is mainly a temporal-resolution problem, not a force-law-learning problem. If smaller substeps are already being used, the most physically transparent choice is to apply the exact Newtonian force directly rather than soften it and then train or use a scalar correction to undo the softening.

4. What should be tested

Version	Zone 2 force rule	Purpose
Current SIMON	$c_{\text{NN}} F_{\text{soft}}$	Existing reported method
No-NN ablation	F_{soft}	Shows effect of leaving force softened
Proposed test	$F_{\text{Newtonian}}$	Tests whether the NN route is unnecessary in Zone 2

Expected outcome: current SIMON and direct Newtonian Zone 2 should be very close, because the current correction is already trying to recover Newtonian force. If direct Newtonian matches or improves the trajectory and energy results, Zone 2 can be simplified without weakening the paper. The revised claim becomes: Zone 2 uses exact Newtonian force plus adaptive sub-stepping; the NN is not needed in this region.

Discussion question for mentor

Is it scientifically better to simplify Zone 2 to direct Newtonian force, given that the current learned correction is analytically equivalent to restoring Newtonian gravity? My view is yes: it removes an unnecessary NN dependency and makes the Zone 2 physics more transparent.